

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF
ENGINEERING) Department of Computer Science and
Engineering**

Course Code: ITA0609

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CO2 ASSESSMENT TOOL 1

Question 1:

1.A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms.

1. Problem Statement

The objective is to maximize the function $f(x) = x^2$ using a Genetic Algorithm (GA). The initial population contains four candidate solutions: 2, 4, 6, and 8. The fitness of each candidate is calculated, the fittest individuals are selected, and crossover is performed by exchanging the least significant bits (LSBs) to produce new offspring.

2. Input

Objective function: $f(x) = x^2$

Initial population: {2, 4, 6, 8}

Selection criterion: Higher fitness value is better

Crossover: Exchange the least significant bits of selected individuals

3. Output

Fitness values of all individuals, the selected best candidates, and new offspring generated through crossover. The role of mutation in maintaining diversity and avoiding premature convergence is also explained.

4. Objective

To maximize $f(x) = x^2$ by selecting individuals with higher fitness and generating improved solutions through genetic operations such as selection, crossover, and mutation.

5. Theory / Introduction

A Genetic Algorithm is an optimization technique inspired by natural selection and evolution. A population of possible solutions is evaluated using a fitness function. The best individuals are selected as parents, and genetic operators such as crossover and mutation are applied to generate new solutions. In this problem, the fitness function is $f(x) = x^2$, so larger values of x produce larger fitness values.

6. Fitness Calculation

The fitness function is:

$$\text{Fitness} = f(x) = x^2$$

Individual (x)	Fitness Calculation	Fitness Value
2	2^2	4
4	4^2	16
6	6^2	36
8	8^2	64

Therefore, the fitness values are:

$$x = 2 \rightarrow \text{Fitness} = 4$$

$x = 4 \rightarrow \text{Fitness} = 16$

$x = 6 \rightarrow \text{Fitness} = 36$

$x = 8 \rightarrow \text{Fitness} = 64$

7. Selection of Best Candidates

Since the objective is maximization, individuals with larger fitness values are preferred. Thus, the individuals are ranked as follows:

Rank	Individual (x)	Fitness
1	8	64
2	6	36
3	4	16
4	2	4

The two best candidates are selected as parents: $x_1 = 8$ and $x_2 = 6$.

8. Binary Representation

For crossover, the selected individuals are represented in 4-bit binary form:

Individual	4-bit Binary Representation
8	1000
6	0110

9. Crossover Operation

The least significant bit (LSB) is the rightmost bit. The selected parents are:

Parent 1: 8 = 1000

Parent 2: 6 = 0110

Exchanging the least significant bits gives:

Parent 1: 100|0

Parent 2: 011|0

In this particular pair, both parents have the same LSB (0). Therefore, exchanging the LSBs produces the same offspring:

Offspring 1: $1000_2 = 8$

Offspring 2: $0110_2 = 6$

Hence, the crossover operation is performed correctly, but no numerical change occurs because both selected parents possess the same least significant bit. This demonstrates that crossover does not always create a different solution.

10. Mutation and Its Role

Mutation is a genetic operation in which one or more bits of an offspring are randomly changed. For binary chromosomes, mutation usually means flipping a bit: 0 becomes 1 or 1 becomes 0.

Example: $1000 \rightarrow 1010$

Here, one bit has changed. This creates a new candidate solution and may introduce genetic material that was not produced through crossover.

11. Advantages of Mutation

Maintains genetic diversity in the population.

Prevents the population from becoming identical too early.

Reduces the possibility of premature convergence to a local optimum.

Introduces new solutions that may have better fitness.

Helps the algorithm explore unexplored regions of the search space.

Improves the overall effectiveness and robustness of the Genetic Algorithm.

12. Flow of the Genetic Algorithm

Initial Population $\rightarrow$ Calculate Fitness $\rightarrow$ Select Fittest Individuals $\rightarrow$ Crossover $\rightarrow$ Mutation $\rightarrow$ Generate New Population $\rightarrow$ Repeat Until a Good/Optimal Solution Is Obtained

13. Conclusion

For the function $f(x) = x^2$, the fitness values of the initial population $\{2, 4, 6, 8\}$ are 4, 16, 36, and 64, respectively. Therefore, $x = 8$ and $x = 6$ are the best candidates. Their binary representations are 1000 and 0110. Exchanging their least significant bits produces the same offspring because both LSBs are 0. Mutation is important because it introduces new genetic variation, maintains diversity, prevents premature convergence, and increases the probability of finding better solutions.

QUESTION 2

2. A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is -0.4 . For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not.

1. Problem Statement

A perceptron is used to classify a customer into one of two classes: purchase or no purchase. The classification is based on advertisement exposure and income level.

2. Input

Advertisement exposure: $x_1 = 2$

Income level: $x_2 = 1$

Weight for advertisement exposure: $w_1 = 0.5$

Weight for income level: $w_2 = 0.3$

Bias: $b = -0.4$

3. Output

The net input of the perceptron and the final binary classification using the step activation function.

4. Perceptron Formula

$$\text{Net Input (z)} = w_1x_1 + w_2x_2 + b$$

5. Step-by-Step Calculation

Substitute the given values:

$$z = (0.5 \times 2) + (0.3 \times 1) + (-0.4)$$

$$z = 1.0 + 0.3 - 0.4$$

$$z = 0.9$$

6. Step Activation Function

The step activation function is defined as:

$$y = 1, \text{ if } z \geq 0$$

$$y = 0, \text{ if } z < 0$$

Since the calculated net input is $z = 0.9$ and $0.9 \geq 0$:

$$y = 1$$

7. Decision

The output of the perceptron is 1. In this classification problem, output 1 represents that the customer will purchase the product.

8. Result Table

Feature	Input (x)	Weight (w)	$w \times x$
Advertisement Exposure	2	0.5	1.0
Income Level	1	0.3	0.3
Bias	—	-0.4	-0.4

$$\text{Net Input} = 1.0 + 0.3 - 0.4 = 0.9$$

$$\text{Step (0.9)} = 1 \rightarrow \text{Purchase}$$

9. Conclusion

The net input to the perceptron is 0.9. Since the net input is greater than or equal to zero, the step activation function produces an output of 1. Therefore, the perceptron predicts that the customer will purchase the product.